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Key Personnel Changes/Previous PI:	None		
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Task Description:	Our proposal is a Flight Definition Study that will use state-of-the-art imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crew members before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises in space so as to establish pain and disc damage mechanisms that will serve as basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crew member injury based on clinical findings. We hypothesize that spontaneous space-flight back pain and disc herniation are due to biomechanical and biological pathomechanisms. First, microgravity leads to higher than normal physiologic disc swelling and increased disc height that may stiffen the lumbar motion segment and cause abnormal segmental movement patterns. These biomechanical changes increase risk for annular rupture, vertebral endplate microfracture, and facet joint capsule strain. Second, increased disc swelling may alter nuclear matrix osmotic pressure and nutrient transport from endplate capillaries in adjacent vertebra. These biological changes adversely affect disc cell metabolism, causing pain and inducing disc matrix degradation. Our project directly addresses the Critical Path Roadmap Risks and Questions for NASA regarding disc injury (Integrated Research Plan (IRP) Gap-B4): Is damage to joint structure, intervertebral discs, or ligaments incurred during or following hypogravity exposure? The goal of this research is to characterize space-flight induced changes comprehensively in disc morphology, biochemistry, metabolism, and kinematics. These data will be correlated with measures of back pain intensity and disability. Crewmembers will be imaged pre-flight to establish baseline data and to characterize measurement repeatability. After long-term microgravity exposure (about 180 days on International Space Station-ISS), crewmembers will be studied while maintaining supine posture as soon as possible after return to 1-G in order to quantify the acute effects of prolonged space flight. Also, pre- and post-flight, they will be imaged supine and in standing upright posture at one body weight and again at 1.1 body weight so that MR images are obtained before and after axial body-weight loads. Pre-flight tests will be done within L-120 to L-60 days before the scheduled space flight. Post-flight tests will be undertaken 1-7 days and 30-60 days after landing to investigate re-adaptation to 1-G and to distinguish immediate and longer-term recoveries. Our proposed measures represent a comprehensive set of tests that evaluate exposure severity, potential injury mechanisms, and pain generator localization. Our research will aid understanding of spinal pain and deconditioning during prolonged microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We propose to use state-of-the-art, non-invasive imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crew members before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises during prolonged microgravity and after re-adaptation to Earth gravity, so as to establish pathomechanisms that will serve as a basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crew member injury based on clinical findings. Importantly, this research will have application to back-pain patients on Earth in general and specifically, to patients exposed to long-term bed rest or lack of mobility (spinal-cord injury patients as well as patients suffering lack of exercise, mobility, and obesity). This research also has application to abnormal spinal curvature and pain suffered by children wearing heavy backpacks to and from school.
	Our IVD (Intervertebral Disc) project was postponed in April 2015 due to reduced availability of ISS crew members for Human Research Program (HRP) research. However, we have made significant progress over the past 12 months by: 1) completing post-flight tests on IVD crewmember 6 and conducting our pre- flight tests on IVD crewmember 7, 2) briefing and consenting ISS crew members, 3) updating Institutional Review Board (IRB) applications, 4) submitting amendments and 5) receiving approvals from the University of California San Diego (UCSD) and NASA-Johnson Space Center (JSC) Institutional Review Boards. Unfortunately, our project was postponed due to loss of crew time by HRP; and we were told to attend and support a joint IVD/In-flight Ultrasound Workshop with the Dulchavsky team at JSC in August 2015. Our IVD project was reinstated in December 2015. Summary of Findings to Date • Lumbar paraspinal muscle atrophy was seen in 5 of 6 crewmembers. The average loss in paraspinal lean muscle mass was 14%. Lumbar paraspinal functional cross-sectional area (FCSA) recovered to 67% by R+45. • Lumbar paraspinal muscle fatigability (using the Biering-Sørensen test) increased post flight, with an average 28% drop in Biering-Sørensen time post flight. • Cervical paraspinal muscle FCSA decreased 17% during long-duration spaceflight (n=4), and recovered only 21% by R+45. • The lumbar curvature changed in all crewmembers, with an average of 11% loss of lumbar spine curvature, some of it due to the vertebrae and some due to the discs. • Some crewmembers showed considerable kyphotic wedging changes in lumbar vertebral bodies, contributing to lumbar spine flattening. • Most crewmembers showed decreased disc wedging that also contributed to lumbar spine flattening. • It is still unclear whether spinal discs swell. On T2 MRI imaging, average disc height did not change from before to after flight, and changes in water content of lumbar discs varied among crewmembers and spinal levels. The investigator team did not observe a systematic increase in disc water content or disc height after spaceflight. • Dynamic fluoroscopy showed that after spaceflight intervertebral movement is decreased in lateral bending and flexion, indicating increased spinal bending stiffness in those directions. 3 of 4 crewmembers had decreased lateral bending and forward flexion over levels L2-L5. In contrast, 4 of 4 crewmembers showed increased extension over levels L2-L5 (testing is done up to 50 degrees flexion and 20 degrees extension). Atrophy of ligaments or muscles would be expected to increase flexibility, but here the spine became stiffer. The investigators hypothesize that loss of lordosis

Task Progress:	moves the spinal segments away from their natural neutral position so as to reduce the range of motion in flexion, while increasing the range of motion in extension. Additionally, stiffness may change because collagen in the ligaments becomes cross-linked due to chronic immobility. Finally, an ASCR staff member commented that the stiffness could be due to protective mechanisms secondary to back pain, because at R+5 crewmembers typically have back pain that causes them to guard their movements. • The straightened spine is more susceptible to disc herniation and other stresses while lifting in a flexed posture. • 3 of 5 crewmembers had postflight pain and/or injury. In one, lumbar disc herniation occurred during ARED (Advanced Resistive Exercise Device) use on ISS and another was noted 30 days post flight. Also, there were numerous endplate irregularities, including before flight. Potential causes for endplate irregularities flight include localized cartilage damage, leading to local inflammation, loss of bone, and vertebral endplate erosions. These can also occur as an inherited abnormality, and thus will be seen in younger individuals and at multiple spinal levels. This finding post flight could be related to MRI observations of vertebral bone marrow edema or lesions. • Literature review by the Principal Investigator team indicates that there is an association between endplate irregularities and low back pain in the general population, that loss of lumbar lordosis increases disc loading, and that lumbar muscle degeneration and fat infiltration (i.e., poor muscle quality) are associated with back pain in the general population. However, it is also well established that there is not always good correlation between anatomical changes, functional deficiencies, and disability. Remaining Research Questions 1. Which spinal levels are at highest risk for disc herniation? In one crewmember who had a postflight herniation at L4-L5, the data suggest that discs adjacent to immobile spinal levels may be at higher risk. 2. Is there a relationship between loss of muscle mass and lumbar spine flattening? For the first 6 crewmembers, a moderate relationship was seen between pre- to postflight changes in muscle mass and spinal curvature. Increasing the sample size will help determine how deconditioning of muscles is affecting lordosis in those with healthy and unhealthy spines. 3. Do established risk factors for terrestrial back pain align with crew injury risk? • Endplate irregularities (weakening), • Bone marrow lesions (inflammation), • Adjacent segment disease, • Trunk muscle atrophy, • Spine flattening IVD Preliminary Major Findings • Decreased functional extensor test (Biering-Sørensen time) • Decreased cervical and lumbar muscle functional cross-sectional area (FCSA) – Partial recovery by R+45 (only 21% in neck) • Increased spine stiffness in flexion • Increased spine straightening • Decreased quality of vertebral bodies ; – Anterior wedging ; – Endplate irregularities • Water content changes are variable • No significant changes to disc height and lumbar length Power analysis shows that the power of the data collected on our first 5 subjects is below 0.8 and that an n=12 (as originally planned for) is still needed. Risk mechanisms for back injury in military personnel are similar to those in returning ISS crews – coming back to gravity is like putting on a heavy backpack. Countermeasure Development for Spinal Health and Prevention of Pain Historically, the countermeasure exercise program was designed to provide maintenance of bone mineral density, aerobic and anaerobic capacity, strength, power, and local muscle endurance. The current crew exercise prescription should be adapted to incorporate elements of spinal rehabilitation. In terrestrial therapy for back pain the goal is restoration of spinal motion segment kinematics as well as the entire kinetic chain through muscle retraining and range-of-motion exercises. Spinal motion segment kinematic treatment involves muscle re-education and manual therapy. Restoration of the kinetic chain involves work on the shoulders, thoracic spine, hips, and knees with an eye toward addressing issues of muscle weakness vs. tightness, and stability vs. mobility of the different joints.
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